BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: DIK-SSC-X-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 15 | **Paper Code:** DIK-SSC-X-2022 (A)

Q1. Choose the correct option:

Q1. If a square matrix A = A^t then A is called a _____ matrix:

(a) Skew symmetric
(b) Symmetric
(c) Rectangular
(d) Null

Q2. When [x−1 4; y+3 −7] = [0 4; −2 −7], then x = ?, y = ?

(a) x = 1, y = −5
(b) x = 1, y = −2/3
(c) x = 1, y = 1
(d) x = 0, y = −2

Q3. √−1 x √−1 =

(A) −2
(B) i
(C) −1
(D) 0

Q4. If log_√3 5 + log_√3 8 − log_√3 2, then a = ?

(A) 13
(B) 10
(C) 30
(D) 20

Q5. Product of $(x-1/x)(x^2+1/x^2)$ = _____

(a) $x^3 - 1/x^3$
(b) $(x - 1/x)^3$
(c) $x^2 + 1/x^3$
(d) $(x + 1/x)^3$

Q6. Zero of polynomial $p(x) = x^2 - 4x + 3$ is _____

(A) 0
(B) 1
(C) −1
(D) 2

Q7. A number of the form $\sqrt[n]{a}$, where a is appositive rational number is called_____

(A) Rational
(B) Irrational
(C) Surd
(D) None

Q8. L.C.M = _____

(a) H / AxB
(b) AxB / H
(c) A / HxB
(d) B / HxA

Q9. Simplified form of $b/(25a^2-b^2) - 1/(5a-b)$ is _____

(a) $5 / 25a^2-b^2$
(b) $+5a / 5a-b$
(c) $-5a / 5a+b$
(d) $-5a / 25a^2-b^2$

Q10. Solution set of $\sqrt{(2x)} + 5 = 9$ is_____

(A) 8
(B) 7
(C) 6
(D) 4

Q11. Let (a,b) be an order of pair in rectangular plane then x-coordinate is called_____

(A) Image
(B) Abscissa
(C) Ordinate
(D) Range

Q12. The line x = a, where a is a real number in parallel to:

(A) y-axis
(B) x-axis
(C) Both A and B
(D) Neither x-axis nor y-axis

Q13. In $\triangle ABC$, if $\angle A \cong \angle B$ then bisector of _____ divides the $\triangle ABC$ into two congruent triangles.

(A) $\angle A$
(B) $\angle B$
(C) $\angle C$
(D) Any one if its angles

Q14. How many equilateral triangles can be made by joining the mid points of the sides of an equilateral triangle?

(A) 4
(B) 3
(C) 2
(D) Cannot be determined

Q15. Which of the following are the sides of a right angle triangle?

(A) 2,3,4
(B) 3,4,5
(C) 4,5,6
(D) 5,6,7

SECTION – B

Time Allowed: 2:40 Hrs | **Paper Code:** DIK-SSC-X-2022 (A)

Q2. Attempt any 9 questions of the following:

(i) Solve the system of linear equations $2x+3y = -1$, $x-y = 2$ with the help of matrices.

(ii) Simplify: $\sqrt{(625x^3y^4/25xy^2)}$

(iii) Simplify: $2.83/(6.52)^3$ with the help of logarithm

(iv) Find the value of $ab+bc+ca$, when $a+b+c = 14$ and $a^2 + b^2 - c^2 = 78$.

(v) Factorize $24x^3 + 3$.

(vi) Find the square root of $x^2 - 2x + 1 + 2xy - 2y + y^2$.

(vii) Find the solution set of $\sqrt{(2x-7)} + 8 = 11$

(viii) Draw the graph of the equation $y - 2x = 6$

(ix) Prove that A(−4, −3), B(1, 4) and C(6, 11) are collinear

(x) The sum of three consecutive odd integers is 81. Find the numbers.

(xi) A line parallel to one side of a triangle, intersecting the other two sides divides them proportionally.

(xii) A ladder whose foot is 2.5m from the front of a house reaches a window 6m above the ground. Calculate the length of a ladder.

SECTION – C

Maximum Marks: 24

> Note: Attempt any THREE questions:

Q3. Prove that points A(−2, −2), B(4, −2), C(4, 6) are the vertices of a right angled triangle.

Q4. If two angles of a triangle are congruent, then the side opposite to these angles are congruent?

Q5. Any point on the right bisectors of a line segment is equidistant from end points of the segment.

Q6. Construct $\triangle xyz$ when $m(\overline{yz}) = 4.1cm$, $m\angle y=60^o$ and $m\angle x=75^o$.